Supplementary Online Content

Shitara K, Van Cutsem E, Bang Y-J, et al. Efficacy and safety of pembrolizumab or pembrolizumab plus chemotherapy vs chemotherapy alone for patients with first-line, advanced gastric cancer: the KEYNOTE-062 phase 3 randomized clinical trial. *JAMA Oncol*. Published online September 3, 2020. doi:10.1001/jamaoncol.2020.3370

eMethods.

eTable 1. KEYNOTE-062 Investigators

eTable 2. Post-discontinuation anticancer therapy in the PD-L1 CPS ≥1 population

eTable 3. Antitumor activity in the intention-to-treat population

eTable 4. Antitumor activity in patients with MSI-H tumors in the PD-L1 CPS ≥1 population

eTable 5. Immune-mediated events and infusion reactions in all treated patients in the PD-L1 CPS ≥1 population

eFigure 1. Type 1 error reallocation strategy

eFigure 2. Overall survival in pre-specified subgroups for pembrolizumab in (A) PD-L1 CPS ≥1 and (B) PD-L1 CPS ≥10 populations and pembrolizumab-chemotherapy versus chemotherapy in (C) PD-L1 CPS ≥1

eFigure 3. Kaplan-Meier estimates of overall survival in patients with (A) MSI-H tumors and (B) without MSI-H tumors in the PD-L1 CPS ≥10 population for pembrolizumab versus chemotherapy

eFigure 4. Kaplan-Meier estimates of progression-free survival for pembrolizumab versus chemotherapy in the (A) PD-L1 CPS ≥1 and (B) PD-L1 CPS ≥10 populations and for pembrolizumab-chemotherapy versus chemotherapy in the (C) PD-L1 CPS ≥1 and (D) CPS ≥10 population

This supplementary material has been provided by the authors to give readers additional information about their work.

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eMethods

Assessments

PD-L1 expression was centrally assessed during screening using the PD-L1 IHC 22C3 pharmDx assay (Agilent Technologies). PD-L1-positive tumors had CPS ≥1 defined as the number of PD-L1-positive cells (tumor cells, macrophages, and lymphocytes) divided by the total number of tumor cells multiplied by 100. Microsatellite instability-high/DNA mismatch repair across 5 mononucleotide repeat markers (*NR21*, *NR24*, *BAT25*, *BAT26*, *MONO27*) was assessed using DNA extracted from formalin-fixed, paraffin-embedded tumor samples and blood (normal control) using the MSI Analysis System, version 1.2 (Promega). Tumors with high levels of microsatellite instability were those with two or more markers changed compared with normal controls.

Endpoints

Changes in health related quality of life assessments from baseline using the EORTC QLQ-C30 and EORTC QLQ-STO22 was a secondary endpoint. Characterization of utilities using the EuroQoL EQ5D-3L was the protocol-specified exploratory endpoint. A full list of exploratory endpoints is found in the protocol and the summary of its amendments. All endpoints were evaluated for pembrolizumab versus chemotherapy and pembrolizumab versus pembrolizumab-chemotherapy in PD-L1 CPS ≥1 and PD-L1 CPS ≥10, except for safety which was evaluated in the as-treated population.

Statistical Analyses

In the final protocol, there were 6 primary hypotheses and one secondary hypothesis: superiority of pembrolizumab-chemotherapy versus chemotherapy for progression-free survival and overall survival in patients with PD-L1 CPS ≥1; superiority of pembrolizumab-chemotherapy for OS in patients with PD-L1 CPS ≥10; non-inferiority of pembrolizumab versus chemotherapy for OS in patients with PD-L1 CPS ≥1; superiority of pembrolizumab versus chemotherapy for OS in patients with PD-L1 CPS ≥1 and PD-L1 CPS ≥10. Superiority of pembrolizumab-chemotherapy versus chemotherapy for ORR in patients with PD-L1 CPS ≥1 was the secondary hypothesis.

The protocol prespecified two efficacy and one safety interim analyses, periodic safety analyses, and a final analysis. The study planned to randomize approximately 750 patients in a 1:1:1 ratio among the 3 treatment groups.
The first efficacy interim analysis was planned to occur ≥10 months after all patients enrolled and at least 317 overall survival events were observed in the pembrolizumab-chemotherapy and chemotherapy arms in patients with PD-L1 CPS ≥1 and used a data cutoff date of September 26, 2018. The data and safety monitoring committee recommended that the study continue to final analysis after reviewing the first interim analysis.
The protocol-specified final analysis was planned for 22 months after randomization of the last patient and accumulation of 415 overall survival events in the pembrolizumab-chemotherapy and chemotherapy arms in patients with PD-L1 CPS ≥1. Analysis of progression-free survival occurred after accumulation of 430 progression-free survival events, the final analysis for progression-free survival occurred at the first interim analysis with a data cutoff date of Sept 26, 2018. This allowed comparison of the superiority of pembrolizumab-chemotherapy versus chemotherapy to have 91% power to show a hazard ratio of 0.65 for progression-free survival or death at a one-sided alpha level of 0.1%, and comparison of the superiority of pembrolizumab-chemotherapy versus chemotherapy, superiority of pembrolizumab versus chemotherapy, and non-inferiority (the statistical criterion for the success of the non-inferiority hypothesis was based on retaining 50% of the effect observed with chemotherapy controls such as cisplatin and 5-fluorouracil versus best supportive care, with the upper bound of the confidence interval for the hazard ratio being less than 1.2) of pembrolizumab versus chemotherapy to have 91%, 82%, and 89% power under a hazard ratio of 0.70, 0.70, and 0.82 for overall survival at a one-sided alpha level of 1.25%, 0.4%, and 0.4%, respectively. Approximately 147 overall survival events had accumulated in the combination therapy and chemotherapy arms and 148 overall survival events had accumulated in the pembrolizumab and chemotherapy arms in patients with CPS ≥10. This allowed comparison of the superiority of pembrolizumab-chemotherapy versus chemotherapy and pembrolizumab versus chemotherapy to have 73% and 63% power to show a hazard ratio of 0.60 and 0.63 for overall survival at a one-sided alpha level of 0.75% and 0.75%, respectively. The data cutoff date for the final analysis was March 26, 2019. This study is registered with ClinicalTrials.gov, number NCT02494583.

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eTable 1. KEYNOTE-062 Investigators

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eTable 2. Post-discontinuation anticancer therapy in the PD-L1 CPS ≥1 population

Concomitant therapies, n (%)	Pembrolizumab N = 254	Pembrolizumab + Chemotherapy N = 250	Chemotherapy N = 244
All patients	134 (53)	118 (47)	132 (54)
Drugs used by >5% of patients in any treatment arm			
Oxaliplatin	90 (35)	19 (8)	22 (9)
Fluorouracil	59 (23)	25 (10)	32 (13)
Paclitaxel	53 (21)	74 (30)	85 (35)
Capecitabine	49(19)	15 (6)	9 (4)
Irinotecan	35 (14)	39 (16)	37 (15)
S-1	34 (13)	8 (3)	5 (2)
Ramucirumab	33 (13)	47 (19)	52 (21)
Cisplatin	26 (10)	8 (3)	6 (3)
Docetaxel	10 (4)	10 (4)	12 (5)
Nivolumab	9 (4)	8 (3)	21 (9)
Pembrolizumab	1 (0.4)	3 (1)	11 (5)
Utomilumab[a]	1 (0.4)	0	0
Atezolizumab[a]	1 (0.4)	0	0
Tislelizumab[a]	0	0	0
Durvalumab[a]	0	0	1 (0.4)

[a]Included although used in <5% of patients to illustrate immunotherapy received post-discontinuation.

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eTable 3. Antitumor activity in the intention-to-treat population

CPS ≥1	Pembrolizumab N = 256	Pembrolizumab + Chemotherapy N = 257	Chemotherapy N = 250
Objective response rate[a], n (%)	**38 (15)**	**125 (49)**	**93 (37)**
Complete response	9 (4)	24 (9)	14 (6)
Partial response	29 (11)	101 (39)	79 (32)
Stable disease	69 (27)	74 (29)	104 (42)
Progressive disease	105 (41)	19 (7)	21 (8)
Not evaluable/no assessment[b]	44 (17)	39 (15)	32 (13)
Median duration of response,c months (95% CI)	13.7 (8.3-21.2)	6.8 (5.5-8.3)	6.8 (5.4-8.2)
Response duration ≥ 12 months, %	51	34	20
CPS ≥10	**Pembrolizumab N = 92**	**Pembrolizumab + Chemotherapy N = 99**	**Chemotherapy N = 90**
Objective response rate,[a] n (%)	23 (25)	52 (53)	34 (38)
Complete response	7 (8)	13 (13)	4 (4)
Partial response	16 (17)	39 (39)	30 (33)
Stable disease	23 (25)	25 (25)	39 (43)
Progressive disease	29 (32)	10 (10)	8 (9)
Not evaluable/no assessment[b]	17 (18)	12 (12)	9 (10)
Median duration of response,[c] months (95% CI)	19.3 (8.3-NR)	8.3 (4.5-29.5)	6.8 (4.8-8.7)
Response duration ≥ 12 months, %	58	44	23

[a]Objective response was defined as confirmed complete or partial response as assessed according to Response Evaluation Criteria in Solid Tumors, version 1.1.
[b]No assessment/non-evaluable captures patients for whom no post-baseline assessments were performed because of death, withdrawal of consent, loss to follow-up or start of new anticancer therapy and patients who had ≥1 post-baseline tumor assessment none of which were evaluable for response determination (e.g. not all target lesions captured).
[c]Duration of response was evaluated in patients who had an objective response and was calculated using the Kaplan-Meier method for censored data.
NR, not reached.

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eTable 4. Antitumor activity in patients with MSI-H tumors in the PD-L1 CPS ≥1 population

	Pembrolizumab N = 14	**Pembrolizumab + Chemotherapy N = 17**	**Chemotherapy N = 19**
Median overall survival, months (95% CI)	Not reached (10.7 to not reached)	Not reached (3.6 to not reached)	8.5 (5.3-20.8)
Median progression-free survival, months (95% CI)	11.2 (1.5 to not reached)	Not reached (3.6 to not reached)	6.6 (4.4-8.3)
Objective response rate,[a] n (%)	8 (57.1)	11 (64.7)	7 (36.8)
Median duration of response,[b] months (95% CI)	21.2 (2.8-NR)	Not reached (2.3-NR)	7.0 (2.0-NR)
Response duration ≥ 12 months, %	71	70	43

[a]Objective response was defined as confirmed complete or partial response as assessed according to Response Evaluation Criteria in Solid Tumors, version 1.1.
[b]Duration of response was evaluated in patients who had an objective response and was calculated using the Kaplan-Meier method for censored data.
NR, not reached.

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eTable 5. Immune-mediated events and infusion reactions in all treated patients in the PD-L1 CPS ≥1 population

Event, n (%)	Pembrolizumab N = 254	Pembrolizumab + Chemotherapy N = 250	Chemotherapy N = 244
Any immune-mediated adverse event[a]	54 (21)	60 (24)	19 (8)
Grade 3-5	15 (6)	14 (6)	4 (2)
Led to discontinuation	4 (2)	11 (4)	2 (1)
Led to death[a]	1 (0.4)	1 (0.4)	1 (0.4)
All immune-mediated adverse events, n (%)			
Hypothyroidism	21 (8)	29 (12)	10 (4)
Hyperthyroidism	11 (4)	11 (4)	0
Pneumonitis	9 (4)	8 (3)	1 (0.4)
Colitis	7 (3)	9 (4)	1 (0.4)
Hepatitis	3 (1)	1 (0.4)	1 (0.4)
Myositis	3 (1)	0	0
Nephritis	3 (1)	1 (0.4)	0
Pancreatitis	2 (1)	0	1 (0.4)
Severe skin reactions	2 (1)	4 (2)	1 (0.4)
Adrenal insufficiency	1 (0.4)	2 (1)	1 (0.4)
Encephalitis	1 (0.4)	0	0
Hypophysitis	1 (0.4)	1 (0.4)	0
Thyroiditis	1 (0.4)	2 (1)	0
Type 1 diabetes mellitus	1 (0.4)	2 (1)	0
Infusion reactions	0	2 (1)	3 (1)

[a]Immune-mediated adverse events and infusion reactions are events of interest based on a list of terms specified by the sponsor regardless of attribution to any study treatment by investigators.
[b]Grade 5 events included pneumonitis in one patient in the pembrolizumab group, enterocolitis in one patient in the pembrolizumab plus chemotherapy group, and pneumonitis in one patient in the chemotherapy group.

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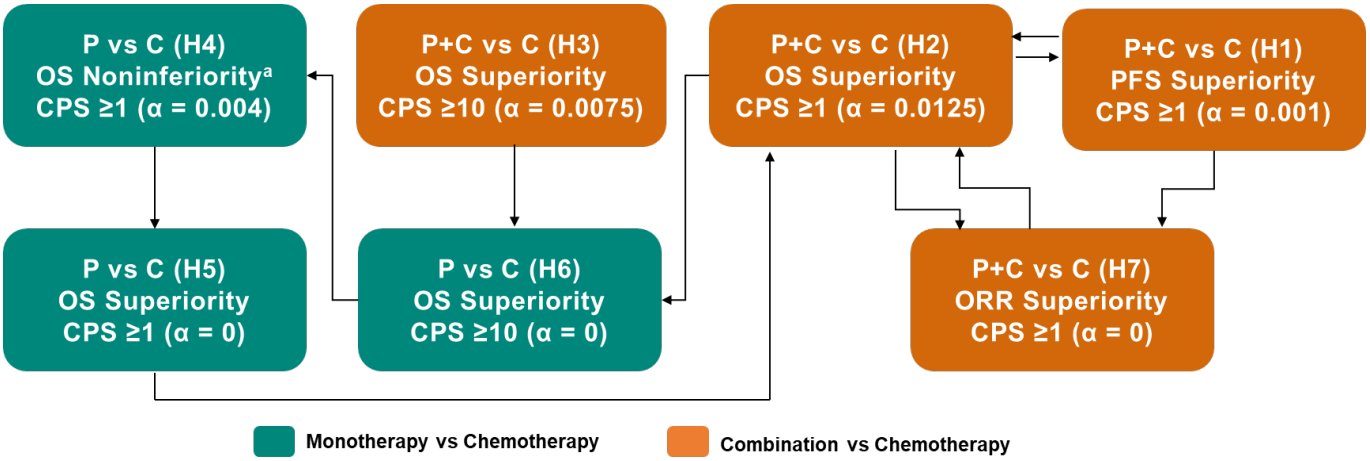
eFigure 1. Type 1 error reallocation strategy



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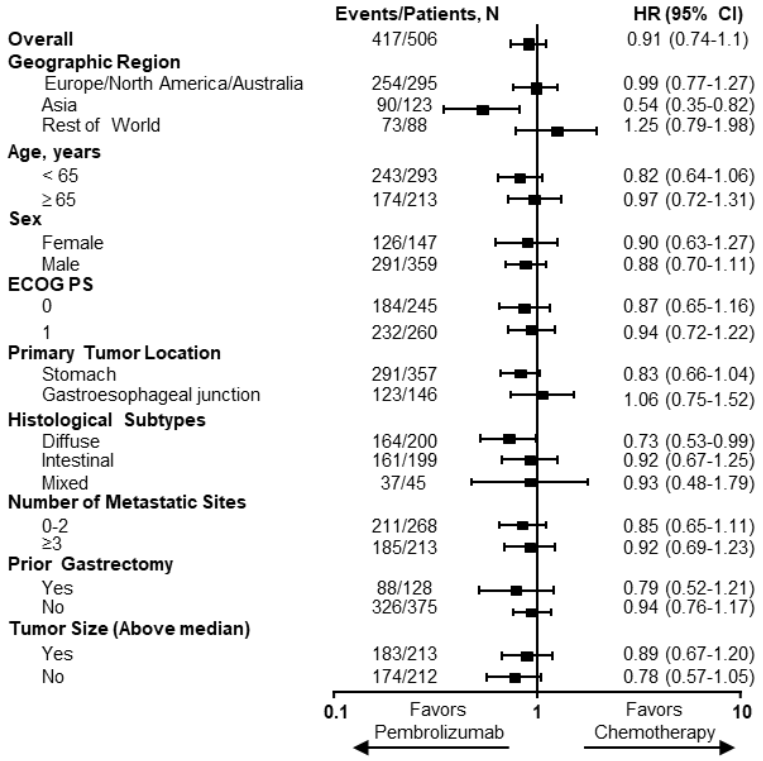
eFigure 2. Overall survival in pre-specified subgroups for pembrolizumab in (A) PD-L1 CPS ≥1 and (B) PD-L1 CPS ≥10 populations and pembrolizumab-chemotherapy versus chemotherapy in (C) PD-L1 CPS ≥1

A



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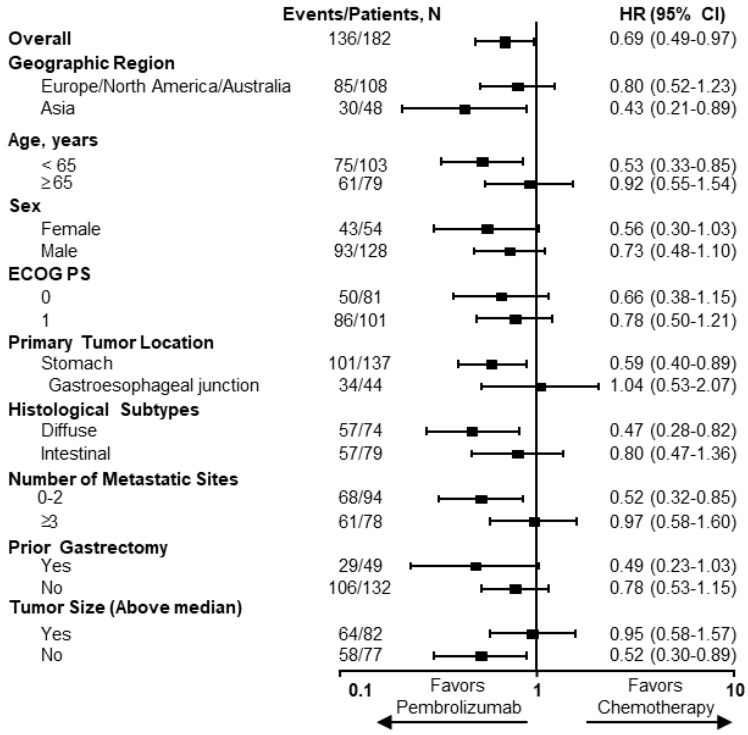

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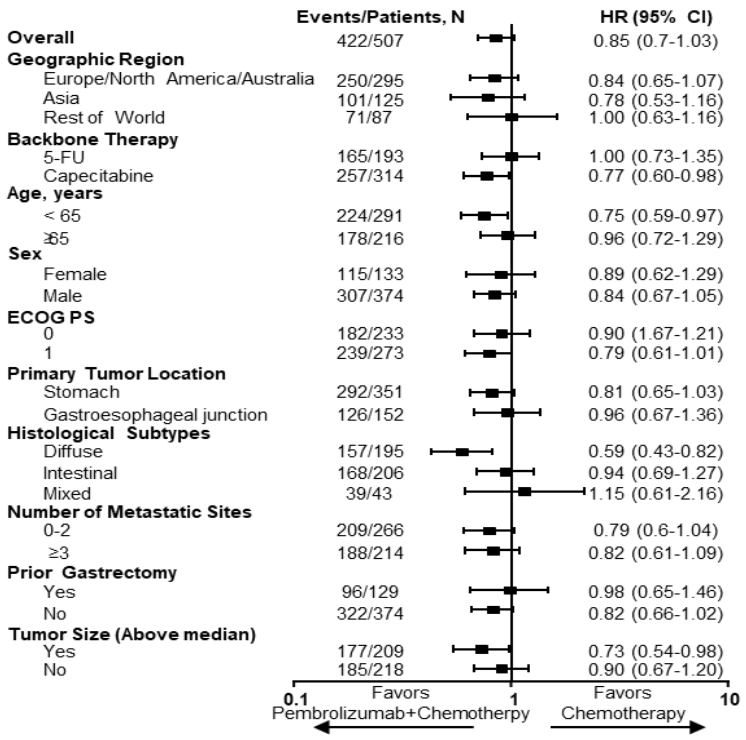
C



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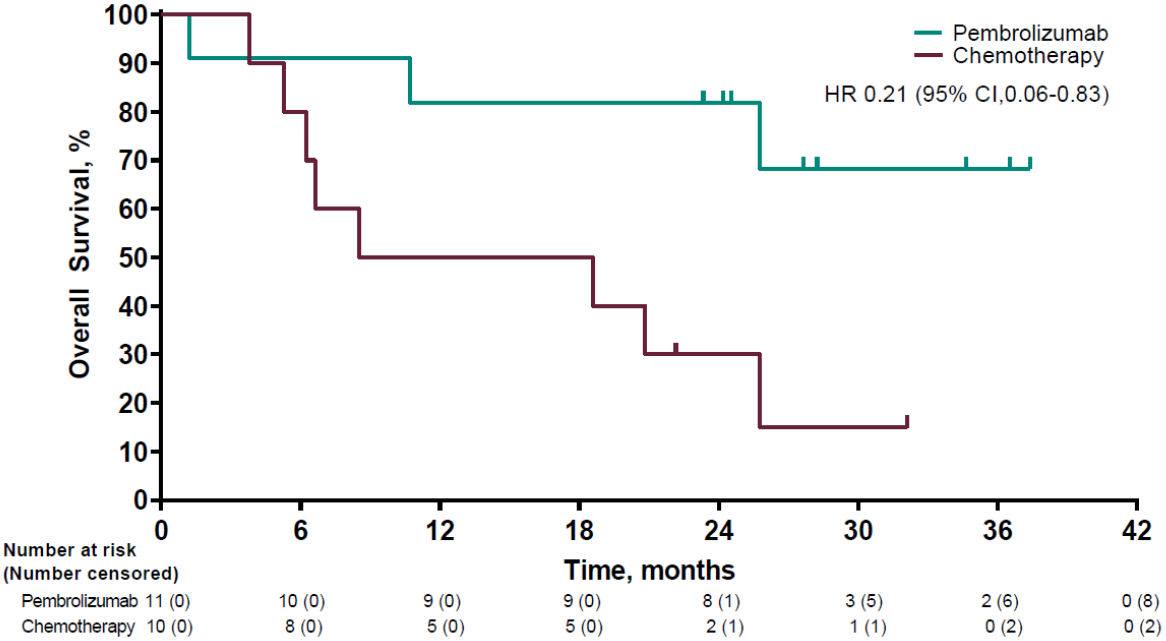
eFigure 3. Kaplan-Meier estimates of overall survival in patients with (A) MSI-H tumors and (B) without MSI-H tumors in the PD-L1 CPS ≥10 population for pembrolizumab versus chemotherapy

A



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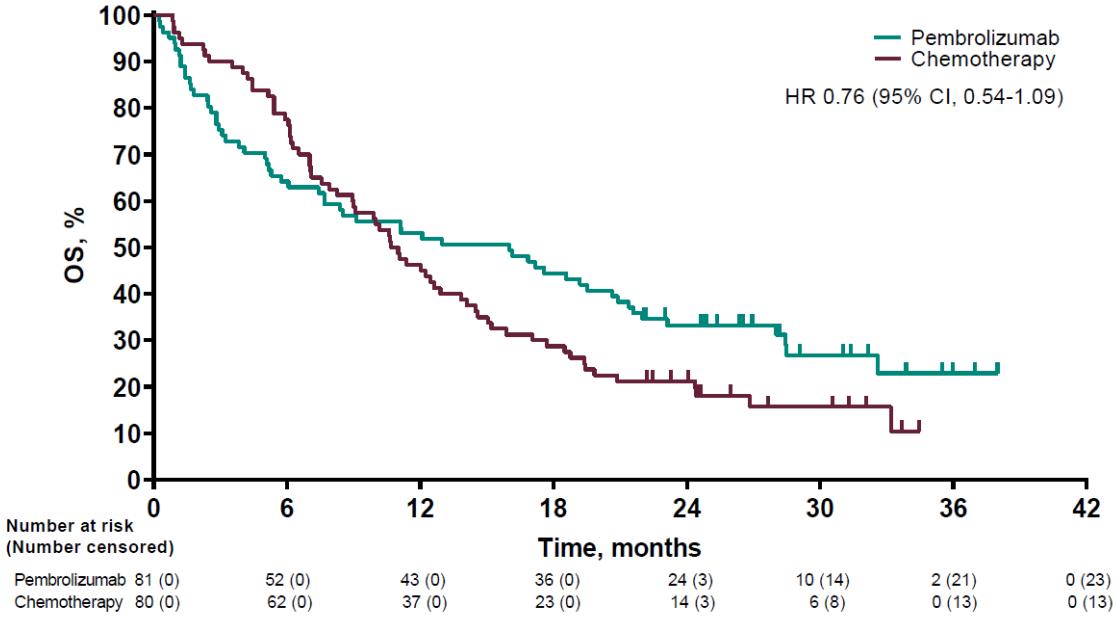

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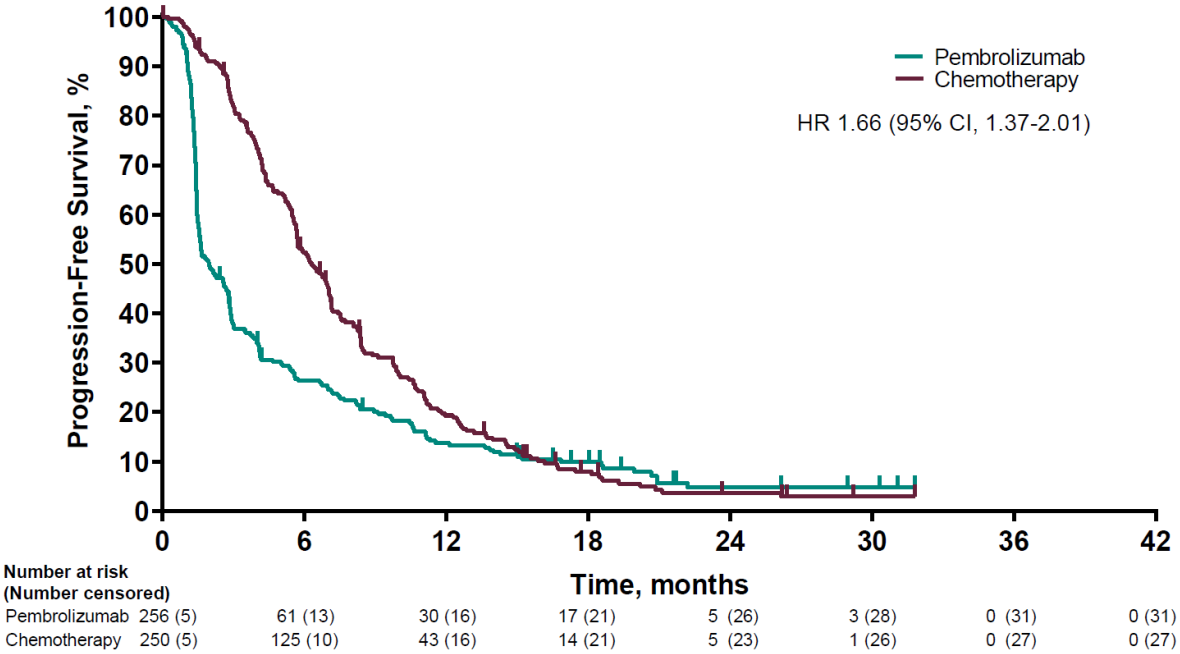
eFigure 4. Kaplan-Meier estimates of progression-free survival for pembrolizumab versus chemotherapy in the (A) PD-L1 CPS ≥1 and (B) PD-L1 CPS ≥10 populations and for pembrolizumab-chemotherapy versus chemotherapy in the (C) PD-L1 CPS ≥1 and (D) CPS ≥10 population

A



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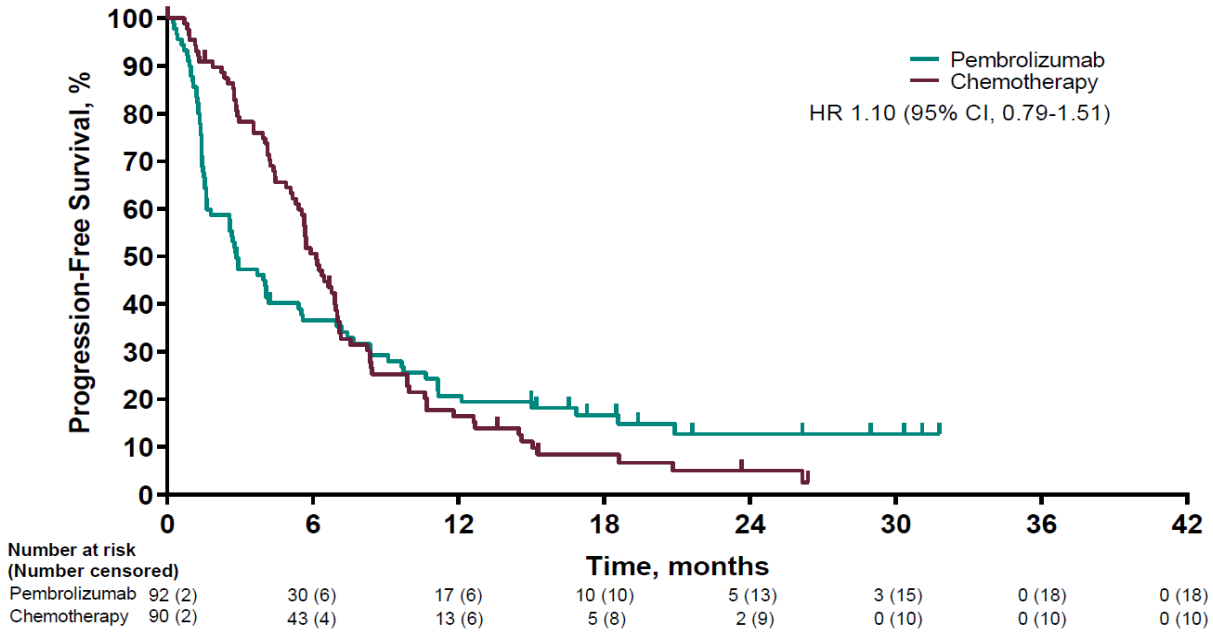

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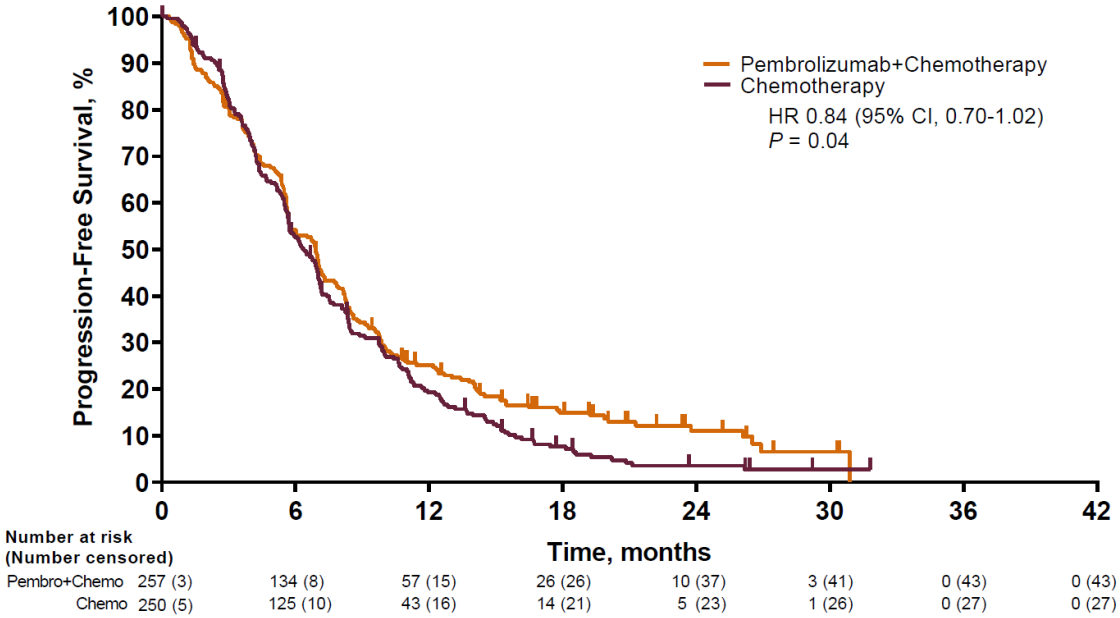
C



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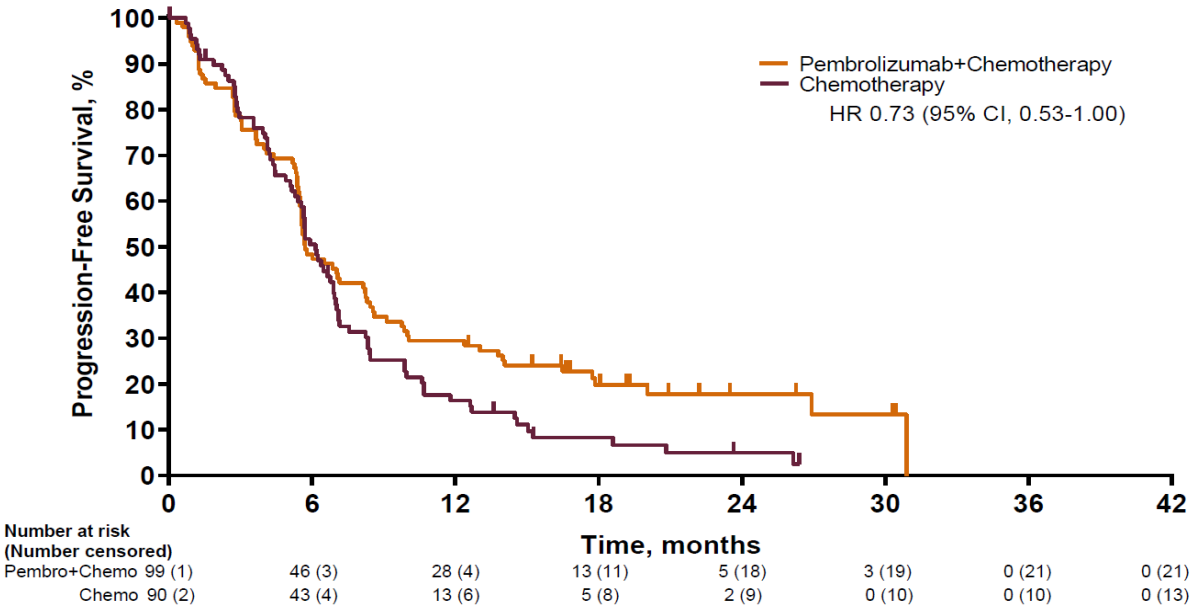

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